

Objects

Kon kitni storage leta ha.	Size		
Refrence Variables	4 bytes / 8 bytes (it varies JVM to JVM) majorly 4 bytes.		
Objects kitne size lete ha. It is divided int 3 parts			
1) Object Headers:	It stores Metadata.	Types of Metadata	Size
		1] Mark words	8 bytes
		2] Class pointer: type of refrence	4 bytes
2) Exact data	Ye vahi data ha jo us object me store ha jse (student data)	Iska size data pr depend krta ha.	
3) Padding	Java me hum koshish krte ha ki har OBJECT (8bytes ke multiple) memory le store hone me	To jitni memory (object Header) or (Exact data) lete ha usko ye nazdiki 8bytes ke multiple tak pahuchane me jitne bytes lagte ha utne add krr deta ha.	

Call By Value

It simply means ki jab hum kisi bhi function ko call krte ha to uske effect main variable pr ni pdta.

Java is a fully call by value language ha.

```
public class CallByValue {
    public static void main(String[] args) {
        // call by value:
        int x = 4;
        int y = 5;
        System.out.println(x + ", " + y); //Result ---> 4, 5
        addTen(x, y);
        System.out.println(x + ", " + y); //Result ---> 4, 5 (kuki addTen ki call values Dynamic hoti ha.)
    }
    static void addTen(int x, int y){
        x = x + 10;
        y = y + 10;
    }
}
```

Deep copy

It fully copy the existing Object to a new Object.

Shallow copy

It points to the existing object or it gives reference to the existing objects.

```
Random r1 = new Random(4, 5);
Random r2 = new Random(r1); // Deep Copy
Random r3 = r1; //Shallow copy.....no "new" keyword.
```